

AIRZEN

AIR INTELLIGENCE & REGULAR FOR POLLUTION ENVIRONMENT NETWORK

Comprehensive System Engineering, Hardware Architecture & Custom Android Application Guide

ESP32 DevKit V1

Custom Android App (APK)

REST API & mDNS

SSD1306 OLED

Automated Healthcare

1. Project Executive Summary

AIRZEN is a multi-purpose, IoT-based smart healthcare and indoor environmental automation station powered by the **ESP32** microcontroller. Operating seamlessly over Wi-Fi, the system continuously analyzes environmental health parameters—hazardous air pollutants, ambient noise levels, temperature, humidity, and fire hazards.

Unlike conventional monitoring setups, AIRZEN provides an integrated autonomous protection loop: it triggers an active **Ventilation Fan** during air pollution or excessive heat, closes an automated **Noise-Containment Window** using a micro servo when sound levels spike, and initiates a **5V Submersible Water Pump** instantly upon flame detection while suppressing ventilation to prevent oxygen feeding the fire. Real-time telemetry is broadcast to an onboard high-tech 0.96" I2C OLED display and synchronized with a dedicated **Custom Android Application (AIRZEN.apk)** featuring audible siren alarms, push notifications, and remote controls for room lighting and door access.

2. Core Functional Subsystems

Subsystem 1: Air Quality Monitoring & Automatic Ventilation (MQ-135)

The MQ-135 electrochemical sensor detects ammonia, benzene, smoke, and hazardous gases. When air quality falls below safe thresholds (ADC reading > 1500), the ESP32 automatically energizes Relay 1 to power on the Ventilation Fan, transmits a high-priority push notification to the user's smartphone, sounds an audible alarm on the phone, and displays live pollution metrics on both the app and OLED display. Built-in hysteresis turns the fan off when clean air levels normalize.

Subsystem 2: Acoustic Control & Automated Window Servo (Sound Sensor)

Ambient noise levels are captured in real-time. When sound spikes beyond the threshold (noise > 2400), **Servo Motor 1 rotates to 90° to close the window**, physically damping external noise. A phone alert and loud alarm notify the occupant. When ambient noise drops into prolonged silence, a low-sound alert is logged. Live sound intensity is streamed in real-time.

Subsystem 3: Climate Regulation (DHT11 / DHT22 Sensor)

Monitors room temperature and humidity. If temperature crosses 35°C, a *"Temperature High"* message is dispatched, the Ventilation Fan turns ON, and the phone sounds an alarm. When humidity exceeds 75%, a *"Humidity High"* alert is triggered; as per design specifications, the ventilation fan is intentionally excluded for humidity.

Subsystem 4: Fire Emergency Suppression (Flame Sensor & 5V Water Pump)

Equipped with an infrared flame sensor. Upon fire detection, Relay 2 immediately activates a **5V Submersible Water Pump** to disperse water throughout the target zone. Concurrently, the system activates a smartphone emergency siren, forces the ventilation fan OFF, and switches the OLED display into an inverted blinking emergency warning.

Subsystem 5: Remote Actuator Controls (Light & Door Servo)

The custom Android app features dual interactive switches: one to toggle room illumination (LED on GPIO 25) and another to command **Servo Motor 2** to lock or unlock the room door between 0° and 90°.

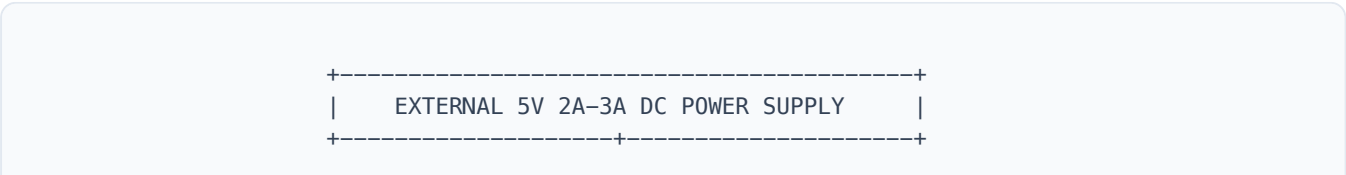
3. Hardware Pinout & Wiring Specification

All sensor pins have been selected to avoid ESP32 ADC2 conflicts with Wi-Fi. Analog inputs (MQ-135 and Sound) use **ADC1** channels (GPIO 34 and GPIO 35), ensuring uninterrupted analog reading while Wi-Fi transmission is active.

Component / Module	Module Pin	ESP32 GPIO	Signal Mode	Power Source
MQ-135 Air Quality Sensor	AOUT (Analog)	GPIO 34	ADC1_CH6 Input	Ext. 5V Rail & Common GND
Sound Detection Sensor	AOUT (Analog)	GPIO 35	ADC1_CH7 Input	3.3V (ESP32) & Common GND
DHT11 / DHT22 Sensor	DATA / OUT	GPIO 4	Digital I/O	3.3V (ESP32) & Common GND
Flame Sensor Module	DOUT (Digital)	GPIO 5	Digital In (Active LOW)	3.3V / 5V & Common GND
0.96" I2C OLED (SSD1306)	SDA / SCL	GPIO 21 / 22	I2C Bus (0x3C)	3.3V (ESP32) & Common GND
Window Micro Servo (SG90)	Signal (Orange)	GPIO 18	PWM (50 Hz)	Ext. 5V Rail & Common GND
Door Micro Servo (SG90)	Signal (Orange)	GPIO 19	PWM (50 Hz)	Ext. 5V Rail & Common GND
Relay 1 (Ventilation Fan)	IN1	GPIO 26	Digital Out (Active LOW)	Ext. 5V Rail & Common GND
Relay 2 (5V Water Pump)	IN2	GPIO 27	Digital Out (Active LOW)	Ext. 5V Rail & Common GND
Room Lighting LED	Anode (+)	GPIO 25	Digital Output	330Ω Resistor to GND
Hardware Buzzer (Optional)	Positive (+)	GPIO 14	Digital / PWM	ESP32 Pin & GND

4. Power Distribution & Common Ground Architecture

CRITICAL POWER RULE: Never power the SG90 servos, relays, and water pump from the ESP32 3.3V or VIN pins! Motor stall currents will cause brownouts and continuous microcontroller reboot cycles. Use an external **5V 2A–3A DC Power Adapter**.



+-----+-----+	
(+5V VCC Rail)	(GND Rail)
+----> SG90 Servo 1 (Window) VCC	+----> Servo 1 GND
+----> SG90 Servo 2 (Door) VCC	+----> Servo 2 GND
+----> 2-Channel 5V Relay VCC	+----> 2-Channel Relay GND
+----> 5V Submersible Pump (via Relay COM-NO)	+----> Water Pump Negative
+----> MQ-135 Gas Sensor VCC	+----> MQ-135 GND
+----> ESP32 VIN (5V Input)	+----> ESP32 GND <=== COMMON

COMMON GROUND REQUIREMENT: The Ground (GND) wire of the external 5V power supply must connect directly to the ESP32 GND pin. Without this common electrical reference, PWM signals to servos and digital control signals to the relay coils will float and fail to switch reliably.

5. Custom Android App (AIRZEN.apk)

AIRZEN operates with a standalone, dedicated Native Android Application (compiled APK size: **5.4 MB**). The application eliminates dependencies on third-party cloud brokers and connects directly to the ESP32 over local Wi-Fi.

Key Mobile App Capabilities

- **Real-Time Visual Dashboard:** Shows live telemetry cards for Air Quality, Sound Amplitude, Temperature, and Humidity.
- **Dynamic Acoustic Siren & Vibration:** The app includes an internal audio synthesis engine (`AlarmSoundManager`) that generates a loud two-tone police emergency siren and triggers phone vibration upon threshold breach.
- **High-Priority Push Notifications:** Utilizes Android's `NotificationChannel.IMPORTANCE_HIGH` to display Heads-Up emergency notifications even when the app is minimized.
- **Interactive Room Controls:** Dedicated Material switches to instantly toggle the Room Light LED and Door Servo lock.
- **Auto-Discovery / Custom IP Config:** Connects automatically to `http://airzen.local` via mDNS or allows manual IP entry.

How to Install & Connect the APK:

1. Locate the compiled `AIRZEN.apk` in your project directory.
2. Transfer `AIRZEN.apk` to your Android phone via USB cable, WhatsApp, Google Drive, or Bluetooth.
3. Tap the file and select **Install** (enable "Allow from this source" if prompted).
4. Connect your phone to the same Wi-Fi network as the ESP32.
5. Open the AIRZEN app, verify the IP shown on the OLED display (or use `airzen.local`), and tap **Connect**.

6. ESP32 Firmware & REST API Endpoints

The ESP32 firmware hosts an embedded HTTP REST server on port 80 with mDNS (`http://airzen.local`) and CORS headers.

Endpoint	Method	Parameters	Response / Action
<code>/api/data</code>	GET	None	Returns full JSON telemetry: air quality, temp, hum, sound, actuator states, and active alert.
<code>/api/control</code>	POST / GET	<code>device=light&state=1 0</code>	Switches Room Light LED ON or OFF.
<code>/api/control</code>	POST / GET	<code>device=door&state=1 0</code>	Rotates Door Servo to 90° (Open) or 0° (Locked).

Endpoint	Method	Parameters	Response / Action
<code>/api/control</code>	POST / GET	<code>device=window&state=1 0</code>	Manual override to open or close window servo.
<code>/</code>	GET	None	Serves the embedded mobile WebApp directly to browsers or WebViews.

7. മലയാളം ചുരുക്കരൂപം (Malayalam Executive Guide)

AIRZEN എന്നത് ESP32-ഉം കസ്റ്റം ആൻഡ്രോയിഡ് ആപ്ലിക്കേഷൻ (APK) ഉപയോഗിച്ച് പ്രവർത്തിക്കുന്ന അത്യാധുനിക ഹെൽത്ത് കെയർ & സ്മാർട്ട് റൂം സിസ്റ്റമാണ്.

◆ 1. എയർ ക്വാളിറ്റി സിസ്റ്റം (MQ-135):

വായുവിൽ പുകയോ വിഷവാതകങ്ങളോ ഉണ്ടായാൽ വെന്റിലേഷൻ ഫാൻ ഓൺ ആകുന്നു. ഒപ്പം ഫോണിലേക്ക് നോട്ടിഫിക്കേഷനും ഫോണിൽ അലാറവും മുഴങ്ങുന്നു. ലൈവ് വാല്യൂസ് ആപ്ലിലും OLED-ലും കാണാം.

◆ 2. സൗണ്ട് കൺട്രോൾ & ജനൽ അടയൽ (Sound Sensor):

റൂമിൽ ശബ്ദം കുടുമ്പോൾ പുറത്തെ ബഹളം ഒഴിവാക്കാൻ മൈക്രോ സെർവോ 1 ഉപയോഗിച്ച് ജനൽ തനിയെ അടയുന്നു. ഒപ്പം ഫോണിൽ അലേർട്ടും അലാറവും ലഭിക്കുന്നു. ശബ്ദം കുറയുമ്പോഴും ഇൻഫർമേഷൻ അലേർട്ട് ലഭിക്കും.

◆ 3. ടെമ്പറേച്ചർ & ഹ്യൂമിഡിറ്റി (DHT11):

റൂമിലെ ചൂട് കുടുമ്പോൾ വെന്റിലേഷൻ ഫാൻ ഓൺ ആകുകയും ഫോണിൽ അലാറം അടിക്കുകയും ചെയ്യുന്നു. ഈർപ്പം കുടുമ്പോഴും അലേർട്ട് ലഭിക്കും (ഇതിൽ ഫാൻ പ്രവർത്തിക്കില്ല).

◆ 4. ഫയർ ഡിറ്റക്ടിംഗ് & വാട്ടർ പമ്പ് (Flame Sensor):

തീ കണ്ടാൽ ഉടൻ തന്നെ 5V വാട്ടർ പമ്പ് പ്രവർത്തിച്ച് വെള്ളം ചിറ്റുന്നു. തീ പടരാതിരിക്കാൻ വെന്റിലേഷൻ ഫാൻ ഓഫ് ആവുകയും ഫോണിൽ ലൗഡ് സൈറൺ അലാറം അടിക്കുകയും ചെയ്യുന്നു.

◆ 5. ആപ്റ്റ് സിട്ലുകൾ (LED ലൈറ്റ് & ഡോർ):

ആപ്ലിലെ ആദ്യത്തെ സിച്ച് വഴി റൂമിലെ ലൈറ്റ് ഓൺ/ഓഫ് ചെയ്യാം. രണ്ടാമത്തെ സിച്ച് വഴി മൈക്രോ സെർവോ 2 പ്രവർത്തിപ്പിച്ച് ഡോർ തുറക്കാനും അടയ്ക്കാനും സാധിക്കും.

◆ 6. OLED ഇൻറർഫേസ് (SSD1306):

പ്രോജക്ട് ഓൺ ചെയ്യുമ്പോൾ "AIRZEN" എന്ന ലോഗോയോട് കൂടി കിടിലൻ ആനിമേഷനും സിസ്റ്റം ചെക്കിംഗും കാണിക്കുന്നു. തുടർന്ന് എല്ലാ റീഡിങ്ങുകളും ക്രമമായി മാറിമാറി കാണിക്കും.

8. System Verification & Testing Checklist

Test Step	Action	Expected Result
Power On Self-Test	Switch on 5V external power supply and ESP32	OLED displays "AIRZEN" circular radar animation, diagnostic steps, and connected IP address.

Test Step	Action	Expected Result
Air Quality Trigger	Introduce smoke or isopropyl alcohol near MQ-135	Relay 1 engages (Fan spins), OLED indicates HAZARDOUS, phone vibrates and sounds audio alarm.
Sound Noise Trigger	Clap loudly near sound sensor	Servo 1 moves to 90° (Window closes), phone pops high noise notification and alarm tone.
Fire Emergency Trigger	Expose flame sensor to a lighter flame	Relay 2 energizes (Water pump sprays water), Fan forced OFF, OLED flashes emergency banner, phone sounds emergency police siren.
Manual Switches	Toggle Room Light and Room Door switches in App	GPIO 25 LED illuminates instantly; Servo 2 smoothly opens and closes door between 0° and 90°.